

Raw phi1 calibration — measured findings

MAIN RESULT

A clear actuator-to-polarization response is demonstrated. Drift is accepted for this objective. Increasing the RP command drives the measured equatorial phase in one direction; reversing the sweep reverses that motion. This result does not require a fitted model or drift correction.

DIRECT EVIDENCE

Endpoint step medians give forward phase changes of -7.547, -7.376 and -7.330 rad over 0–0.8 V, each exceeding one full turn. Reverse sweeps give +7.430, +8.465 and +7.098 rad. Internal OUT1 medians agree with the requested command within 0.12 mV.

SCOPE AND STOP STATUS

18 September 2026. Three complete forward/reverse cycles, one completed fixed-voltage hold, and one finer scan interrupted at the user request. Both outputs returned to zero; saved digital cleanup traces confirm zero. No further acquisition was started.

QUALIFICATION: TRAJECTORY GEOMETRY

S1 ranges from -0.0394 to 0.1942 in the repeat scan. Its departure is organized by measured phase across cycles. Three-second settling also gives nonzero S1, up to 0.2091. The response therefore demonstrates actuation without establishing an exact $S1=0$ circle.

DRIFT CONTEXT

At fixed 0.4 V, phase spans 0.419 rad over 74.4 s; the first-to-last block medians change by -0.188 rad. Internal OUT1 remains at 0.39990234375 V. This variation is much smaller than the >7 rad commanded full-sweep excursion, although durations differ and this is not a noise-subtracted comparison.

VOLTAGE INTERPRETATION

The following page brackets a π change in $\text{atan2}(S3, -S2)$ using measured RP command steps. Internal FPGA readings are not electrical BNC or terminal measurements: the expected 13 V terminal V_{π} remains unverified. No fitted calibration or assumed driver gain is applied. Actual PAX port and physical IN1/IN2 connections await confirmation.

Observed equatorial π -change brackets

All intervals are ΔRP requested command volts relative to the stated pass reference.

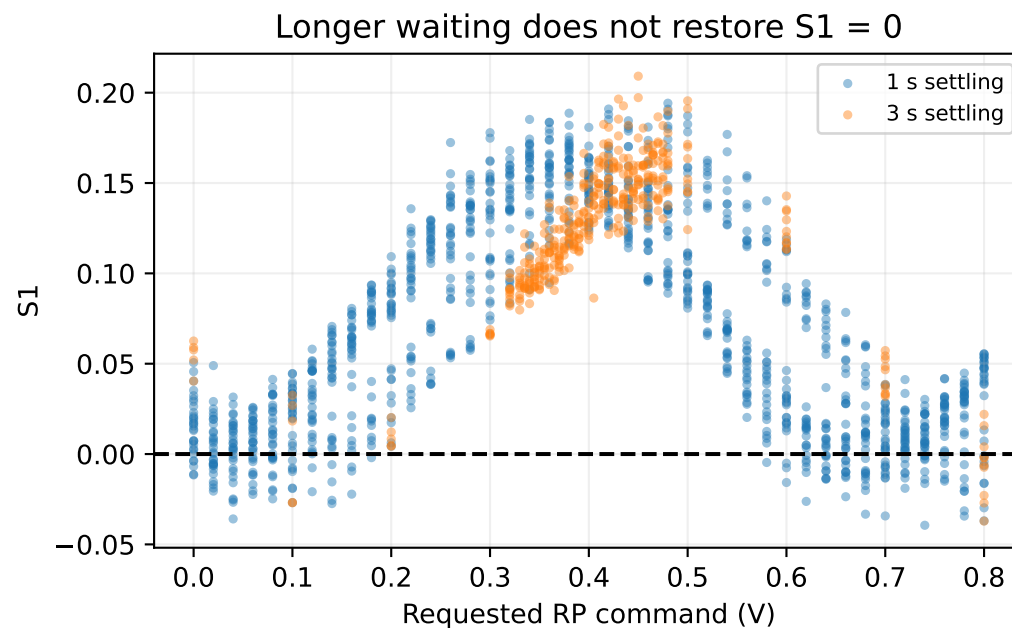
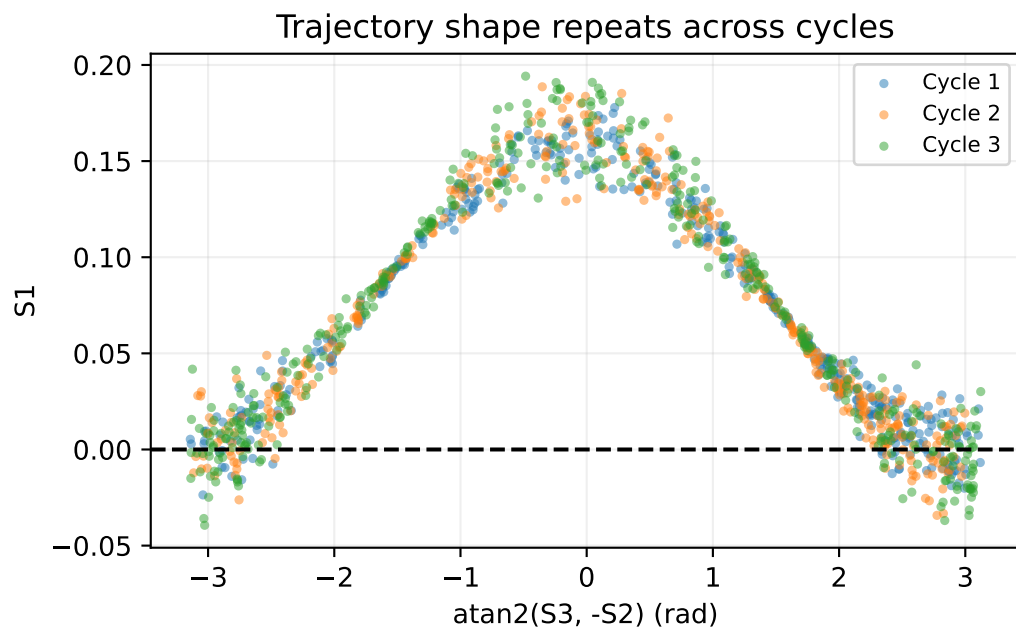
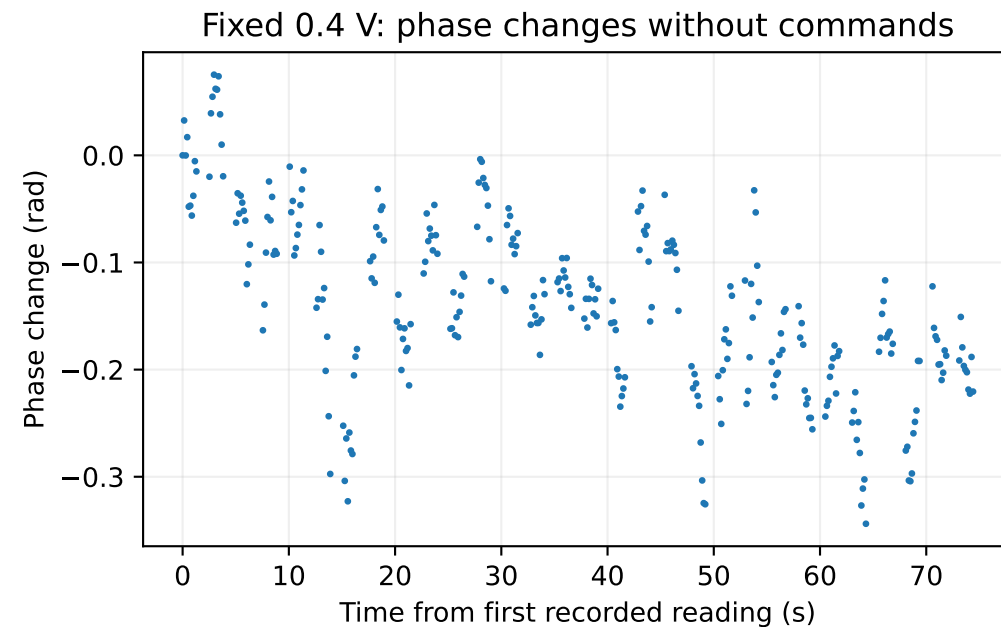
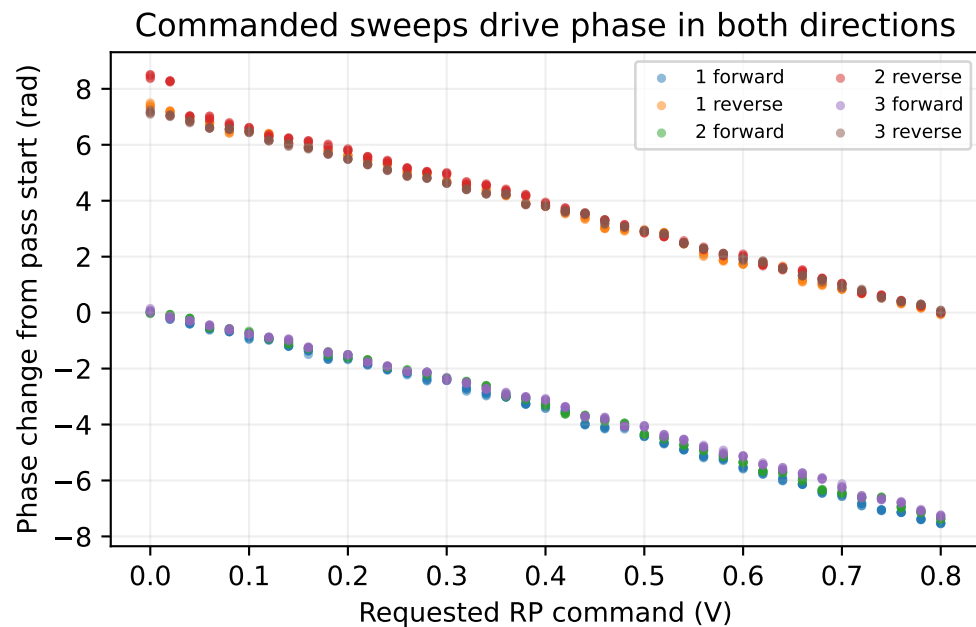
Settle	Cycle	Direction	Reference V	Step-median bracket	Sample-spread bracket
1 s	1	forward	0.0	0.360–0.380	0.360–0.380
1 s	1	reverse	0.8	0.340–0.360	0.340–0.360
1 s	2	forward	0.0	0.380–0.400	0.380–0.400
1 s	2	reverse	0.8	0.320–0.340	0.300–0.340
1 s	3	forward	0.0	0.380–0.400	0.360–0.420
1 s	3	reverse	0.8	0.320–0.340	0.320–0.360
3 s	1	forward	0.0	0.350–0.355	0.350–0.360
3 s	1	reverse	0.8	0.325–0.330	0.320–0.330

Step-median brackets use adjacent measured levels. Wider brackets include all observed reference/endpoint samples.

No interpolation. The sample-spread bracket is an observed range, not a statistical confidence interval.

Different grids and timing expose drift as well as settling; the 3 s run is not an isolated settling-time experiment.

Demonstrated actuator response — raw measurements



Every dot is a recorded PAX reading. Original axes are retained; no fitted rotation, drift removal or smoothing.

Active parameters and remaining measurements

195031_RAW-PHI1_REPEATABILITY-1S

Status: completed. 0.0–0.8 V, 0.02 V steps

1.0 s wait; 5 readings/block; 3 cycles (ignored in hold mode). 1230 raw readings, 1230 unique PAX timestamps.

Full settings, exact raw records, one consolidated scope.npz and individual PDF are in this source folder.

200038_RAW-PHI1_FIXED-0P4-DRIFT

Status: completed. Fixed 0.4 V, 30 blocks

1.0 s wait; 10 readings/block; 3 cycles (ignored in hold mode). 300 raw readings, 300 unique PAX timestamps.

Full settings, exact raw records, one consolidated scope.npz and individual PDF are in this source folder.

200242_RAW-PHI1_FINE-BRACKETS-3S

Status: interrupted. Explicit ascending levels: 0,0.1,0.2,0.3,0.32,0.33,0.335,0.34,0.345,0.35,0.355,0.36,0.365,0.37,0.375,0.38,0.385,0.39,0.395,0.4,0.405,0.41,0.415,0.42,0.425,0.43,0.435,0.44,0.445,0.45,0.455,0.46,0.465,0.47,0.475,0.48,0.5,0.6,0.7,0.8

3.0 s wait; 5 readings/block; 1 cycles (ignored in hold mode). 381 raw readings, 381 unique PAX timestamps.

Full settings, exact raw records, one consolidated scope.npz and individual PDF are in this source folder.

PAX ACQUISITION

Configured wavelength 830.0 nm; mode 5; rotation 60.0 Hz. Orientation Stokes are normalized from measured angles; DOP is preserved separately.

NEXT DISCRIMINATING MEASUREMENTS

Confirm PAX port and physical input connections. Measure RP BNC/MDT input and piezo terminals with an independent, appropriately rated instrument while repeating known held commands. Compare their time records with PAX. Once voltage stability is established, check optical state preparation and the PAX reference axes to identify the systematic S1 departure.

INTERPRETATION LIMIT

These tests characterize static held commands. A reliable sinusoidal trajectory is not established by these scans; it needs a time-resolved test after the mapping and drift are understood.